Campus Recruitment Management System

(CRMS) System Architecture, Module Breakdown & Code Guide

Project: CRMS (Mini Project)	**Stack:** PHP 8.2 (MVC) + MySQL / MariaDB
Frontend: Bootstrap 5.3 + Vanilla CSS	**Server:** Apache (XAMPP) / Port 80 & 3306
Database: crms_db (6 Relational Tables)	**Auth:** Role-based (Admin, Coordinator, Student)

1. Configuration & Database Layer (config/)

The foundation of CRMS consists of **config.php** and **db.php**. These handle security initialization, environment detection, and the PDO database connection singleton.

A. Global Config & Security (config/config.php)

• **Session & Cookie Sync:** Implements secure HMAC-SHA256 signed session cookie tracking.
• **CSRF Token Defense:** Generates a 32-byte cryptographic token via generateCsrfToken() embedded into every form via csrfField(). Prevents cross-site attacks.
• **XSS Protection & Sanitization:** The e() helper escapes user inputs using htmlspecialchars() to neutralize malicious script injections.
• **Flash Notification System:** Implements one-time alert toasts (e.g., 'Application submitted successfully').

B. Database Manager & Auto-Migration (config/db.php)

The Database class uses the **Singleton Pattern**. If the database crms_db does not exist, the system creates it on the fly and automatically imports database/schema.sql and database/seed.sql.

```
// Example: PDO Singleton with Auto-Migration in config/db.php
$dsn = "mysql:host={$host};port={$port};dbname={$dbName};charset=utf8mb4";
$options = [
  PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
  PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC
];
$pdo = new PDO($dsn, $username, $password, $options);
```

2. Authentication & Role-Based Authorization

CRMS enforces a strict separation between **Students** and **Placement Administrators**. Controlled by AuthController.php and User.php.

User Role	Authentication Table	Landing Dashboard	Privileges & Capabilities

Super Admin	admins (role='superadmin')	?page=admin/dashboard	Full control over students, companies, jobs, offers, and analytics reports.
Coordinator	admins (role='coordinator')	?page=admin/dashboard	Manage job drives, shortlist applications, and update interview selections.
Student	students	?page=student/dashboard	Manage academic profile, browse eligible placement drives, and track job offers.

Password Security Example: All passwords use standard bcrypt algorithm (password_hash($pass, PASSWORD_BCRYPT)). During login, password_verify($inputPass, $row['password_hash']) validates credentials without storing plaintext.

3. Student Module & Automated Eligibility Engine

The student module allows eligible candidates to view placement opportunities, apply with a single click, and track interview progress.

Automated Eligibility Engine (app/models/Job.php)

Before a student can click 'Apply', the backend dynamically validates four criteria:
1. **CGPA Cutoff:** Verifies if student.cgpa >= job.min_cgpa.
2. **Degree / Course Match:** Checks if student's course (e.g. 'B.Tech CSE') is inside job.allowed_courses.
3. **Current Semester:** Checks if student's semester is listed in job.allowed_semesters (e.g. '7,8').
4. **Skill Match Percentage:** Calculates overlap between student skills and job requirements (e.g. Python, SQL).

Concrete Example:
• **Job Drive:** Google Software Engineer | Min CGPA: **7.50** | Allowed Courses: **B.Tech CSE, B.Tech IT**
• **Candidate A (Rahul):** CGPA: **8.45** | Course: **B.Tech CSE** → **ELIGIBLE** (Can Click 'Apply')
• **Candidate B (Priya):** CGPA: **7.10** | Course: **B.Tech CSE** → **INELIGIBLE** ('CGPA cutoff of 7.50 not satisfied')
• **Candidate C (Amit):** CGPA: **8.90** | Course: **MCA** → **INELIGIBLE** ('Course MCA not listed in allowed degrees')

4. Placement Administrator Module & Screening Pipeline

Managed through AdminController.php. Provides full end-to-end recruitment management:

• **Company Directory:** Add, edit, and maintain partner recruiting companies, industry type, and contact person.
• **Job Openings & Placement Drives:** Schedule drives, CTC packages, eligibility cutoffs, application deadlines, and drive status (Open / Closed).
• **Application Screening Pipeline:** Filter candidates by CGPA, skills, or status. Step-by-step state progression:
  Applied → Under Review → Shortlisted → Selected / Rejected
• **Offer Management:** When candidate is marked **'Selected'**, the system records an official **Offer Reference ID** (e.g. GOOG-2026-OFF-019) in the selections table.
• **Reports & Analytics:** Aggregated placement statistics, total offers, average salary package, and department-wise charts.

5. Database Relational Design (crms_db)

The relational database schema is normalized and enforced with Foreign Keys and Cascading Rules:

Table	Primary Key	Foreign Keys	Description & Purpose
admins	admin_id	None	System administrators and placement coordinators.
students	student_id	None	Registered students, academic profile (CGPA, Course, Skills, Resume).
companies	company_id	None	Partner recruiting corporate entities and contacts.

jobs	job_id	company_id → companies	Placement drive postings, salary package, min CGPA, deadlines.
applications	application_id	job_id, student_id	Tracks student applications. UNIQUE constraint on (job_id, student_id).
selections	selection_id	application_id → applications	Final interview outcome, official offer letter reference & remarks.

6. Quick Start, Testing & Seeded Credentials

Use these verified credentials to test each portal of the application locally:

Portal	Email Address	Password	Role / Permissions
Admin Portal	admin@crms.edu	Admin@123	Super Admin (Full Placement & Student Access)
Coordinator Portal	coordinator@crms.edu	Coord@123	Placement Officer (Drive & Application Screening)
Student Portal	rahul.sharma@crms.edu	Student@123	Candidate (B.Tech CSE, CGPA: 8.45)
Student Portal	priya.patel@crms.edu	Student@123	Candidate (B.Tech IT, CGPA: 7.90)

Execution Endpoints:

- **Application Web Portal:** http://localhost/crms/
- **phpMyAdmin Database Explorer:** http://localhost/phpmyadmin/
- **Direct Login URL:** http://localhost/crms/public/index.php?page=login

Document Summary: This document serves as the complete academic technical specification for CRMS. It covers the software architecture, database entities, security models, eligibility validation engine, and administrative workflows.